

aerofit-data-exploration

March 19, 2024

```
[3]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
[4]: df = pd.read_csv('aerofit_data.csv')
df.head()
```

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
0	KP281	18	Male	14	Single	3	4	29562	112
1	KP281	19	Male	15	Single	2	3	31836	75
2	KP281	19	Female	14	Partnered	4	3	30699	66
3	KP281	19	Male	12	Single	3	3	32973	85
4	KP281	20	Male	13	Partnered	4	2	35247	47

Analyzing Basic Metrics: Observations on the Shape of Data, Data Types, Conversion of Categorical Attributes to ‘Category’, Missing Value Detection, Statistical Summary

```
[5]: # listing all the column names
print("Shape IS : ", df.shape)
print()
print("Column names",df.columns)
print("Data type")
print(df.dtypes)
```

Shape IS : (180, 9)

```
Column names Index(['Product', 'Age', 'Gender', 'Education', 'MaritalStatus',
'Usage',
'Fitness', 'Income', 'Miles'],
dtype='object')
```

Data type

Product	object
Age	int64
Gender	object
Education	int64
MaritalStatus	object
Usage	int64

```

Fitness          int64
Income           int64
Miles            int64
dtype: object

```

```
[6]: df.describe()
```

```

[6]:
      count    Age  Education  Usage  Fitness  Income \
count  180.000000  180.000000  180.000000  180.000000  180.000000
mean    28.788889   15.572222   3.455556   3.311111  53719.577778
std      6.943498    1.617055   1.084797   0.958869  16506.684226
min     18.000000   12.000000   2.000000   1.000000  29562.000000
25%     24.000000   14.000000   3.000000   3.000000  44058.750000
50%     26.000000   16.000000   3.000000   3.000000  50596.500000
75%     33.000000   16.000000   4.000000   4.000000  58668.000000
max     50.000000   21.000000   7.000000   5.000000 104581.000000

      count    Miles
count  180.000000
mean    103.194444
std      51.863605
min     21.000000
25%     66.000000
50%     94.000000
75%    114.750000
max     360.000000

```

```

[7]: # Check for unique values and value counts for categorical attributes
for col in df.select_dtypes(include='object'):
    print(col, ":", df[col].unique())
    print(df[col].value_counts())

```

```

Product : ['KP281' 'KP481' 'KP781']
KP281    80
KP481    60
KP781    40
Name: Product, dtype: int64
Gender : ['Male' 'Female']
Male      104
Female     76
Name: Gender, dtype: int64
MaritalStatus : ['Single' 'Partnered']
Partnered  107
Single      73
Name: MaritalStatus, dtype: int64

```

```
[8]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 180 entries, 0 to 179
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Product               180 non-null   object
1   Age                   180 non-null   int64
2   Gender                180 non-null   object
3   Education              180 non-null   int64
4   MaritalStatus         180 non-null   object
5   Usage                 180 non-null   int64
6   Fitness               180 non-null   int64
7   Income                180 non-null   int64
8   Miles                 180 non-null   int64
dtypes: int64(6), object(3)
memory usage: 12.8+ KB

```

```

[9]: # analyzing duplicate records
df[df.duplicated()]

# df.drop_duplicates(inplace=True)

```

```

[9]: Empty DataFrame
Columns: [Product, Age, Gender, Education, MaritalStatus, Usage, Fitness,
Income, Miles]
Index: []

```

```

[10]: # analyzing the null values
df.isnull().sum().sort_values(ascending=False)

```

```

[10]: Product      0
Age              0
Gender           0
Education        0
MaritalStatus    0
Usage            0
Fitness          0
Income           0
Miles            0
dtype: int64

```

```

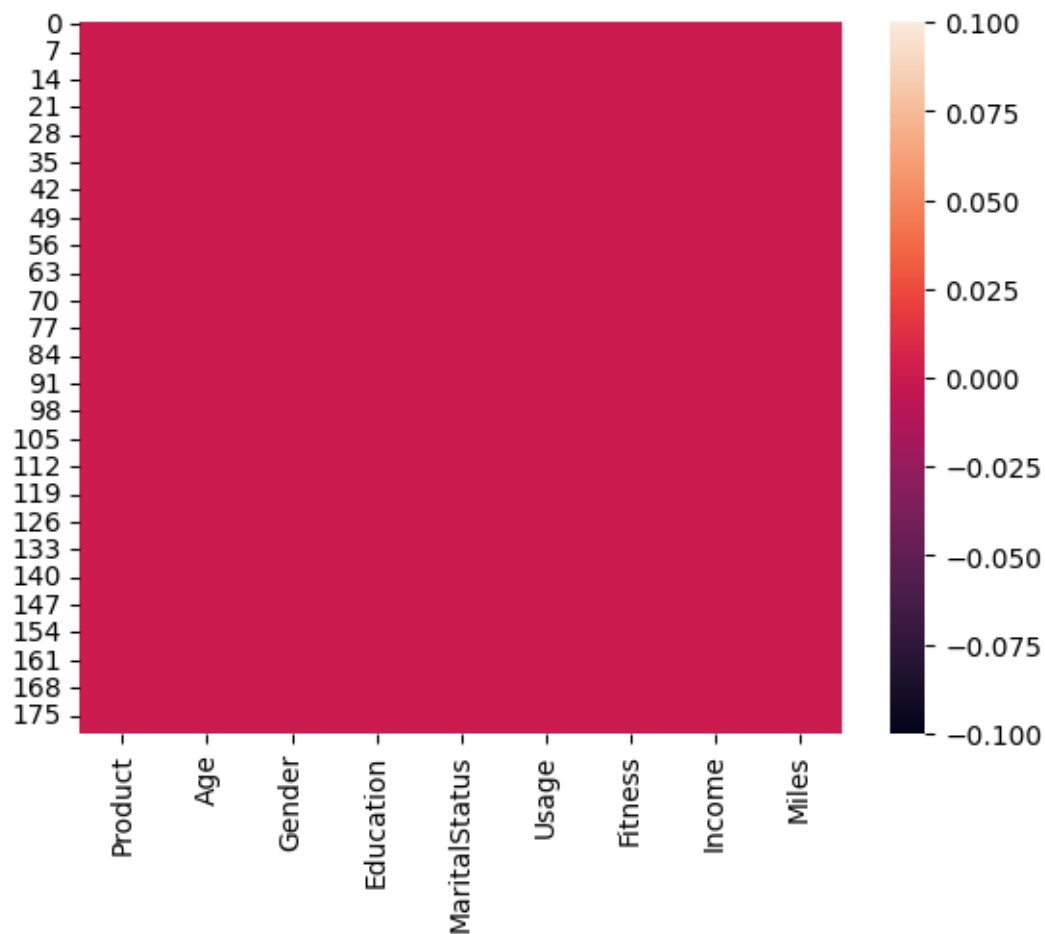
[11]: sns.heatmap(df.isnull())

```

```

[11]: <Axes: >

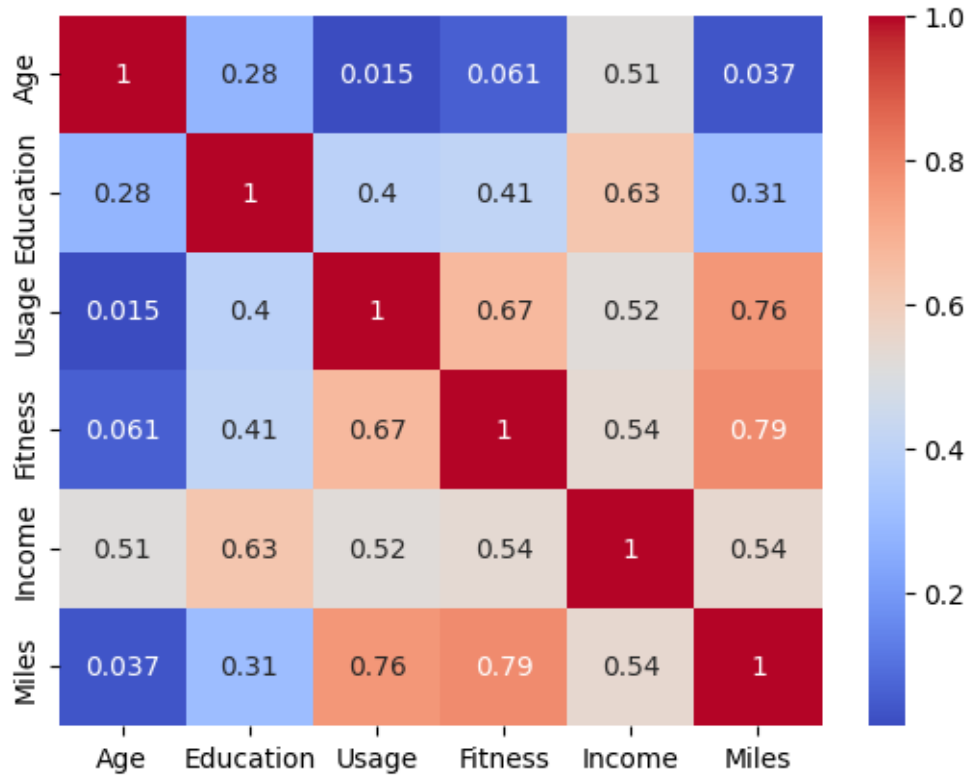
```



```
[20]: # Heatmap for correlation
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
plt.show()
```

<ipython-input-20-a788518339b5>:2: FutureWarning: The default value of numeric_only in DataFrame.corr is deprecated. In a future version, it will default to False. Select only valid columns or specify the value of numeric_only to silence this warning.

```
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
```



In the above heatmap linear relationship between data points is evaluated

Correlation between Age and Miles is 0.03

Correlation between Education and Income is 0.62

Correlation between Usage and Fitness is 0.66

Correlation between Fitness and Age is 0.06

Correlation between Income and Usage is 0.51

Correlation between Miles and Age is 0.03

1 Business Insights

Age Outliers:

Products KP281 and KP781 each have 2 outliers, while KP481 has 1 outlier. These outliers represent customers purchasing treadmills at ages significantly different from the typical customer age range.

Income Outliers:

Product KP781 has 19 outliers in income, indicating a considerable number of high-income customers. Targeted marketing and premium offerings can capitalize on this segment.

Usage Outliers:

Product KP781 has 9 outliers in usage, indicating a subset of customers who use the treadmill much more frequently than the average user, possibly for intensive fitness regimens or training.

Fitness Outliers:

KP281 and KP481 each have 1 outlier in fitness level, suggesting a few customers with significantly different fitness levels. Understanding their preferences can inform product development and marketing strategies.

Miles Outliers:

Product KP781 has 11 outliers in miles logged, indicating customers who consistently log significantly more miles than average. Understanding their motivations can help tailor product features and support long-distance running.

Non-Graphical Analysis

```
[12]: # Value counts and unique attributes
print("Value Counts:")
print(df['Product'].value_counts())
print(df['Gender'].value_counts())
print(df['MaritalStatus'].value_counts())

# Unique attributes
print("Unique attributes:")
print(df['Age'].unique())
print(df['Education'].unique())
print(df['Usage'].unique())
print(df['Income'].unique())
print(df['Fitness'].unique())
print(df['Miles'].unique())
```

Value Counts:

KP281 80

KP481 60

KP781 40

Name: Product, dtype: int64

Male 104

Female 76

Name: Gender, dtype: int64

Partnered 107

Single 73

Name: MaritalStatus, dtype: int64

Unique attributes:

[18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41

43 44 46 47 50 45 48 42]

[14 15 12 13 16 18 20 21]

[3 2 4 5 6 7]

[29562 31836 30699 32973 35247 37521 36384 38658 40932 34110

39795	42069	44343	45480	46617	48891	53439	43206	52302	51165								
50028	54576	68220	55713	60261	67083	56850	59124	61398	57987								
64809	47754	65220	62535	48658	54781	48556	58516	53536	61006								
57271	52291	49801	62251	64741	70966	75946	74701	69721	83416								
88396	90886	92131	77191	52290	85906	103336	99601	89641	95866								
104581	95508]																
[4	3	2	1	5]													
[112	75	66	85	47	141	103	94	113	38	188	56	132	169	64	53	106	95
212	42	127	74	170	21	120	200	140	100	80	160	180	240	150	300	280	260
360]																	

2 Business Insights:

Product Popularity:

The entry-level treadmill model, KP281, has the highest sales volume among the three models, indicating its popularity among customers. This suggests that there is a significant demand for affordable treadmill options in the market.

Customer Demographics:

The customer base consists of a slightly higher number of male customers compared to female customers. This could indicate potential opportunities to target marketing efforts towards male customers or to explore ways to appeal more to female customers. Partnered individuals make up the majority of customers, suggesting that AeroFit's products may appeal to individuals looking for fitness solutions for their families or households.

Market Segmentation:

The age range of customers spans from young adults to middle-aged individuals, indicating a broad appeal across different age groups. This highlights the need for diverse product offerings and marketing strategies to cater to varying age demographics. Income levels vary widely among customers, suggesting the importance of offering products at different price points to accommodate customers with varying budgets.

Customer Expectations:

Customers have diverse expectations regarding treadmill usage and fitness goals, as indicated by the range of expected weekly usage and miles to be walked/run. This underscores the importance of offering versatile treadmill features to meet the needs of customers with different fitness objectives.

Potential Growth Areas:

There may be opportunities to further penetrate the market by introducing more mid-level treadmill options or expanding the features of existing models to cater to specific customer preferences. Targeted marketing campaigns and promotions can capitalize on the popularity of the entry-level model while also highlighting the unique features of higher-end models to attract customers looking for more advanced fitness solutions.

Customer Engagement and Satisfaction:

Personalized customer engagement initiatives, such as tailored fitness plans or loyalty programs, can enhance customer satisfaction and foster long-term customer loyalty. Gathering feedback from

customers across different demographics can provide valuable insights for product improvement and customer service enhancements, ultimately driving customer satisfaction and retention.

3 *Univariate Visual Analysis*

```
[13]: sns.distplot(df['Age'])  
      plt.show()
```

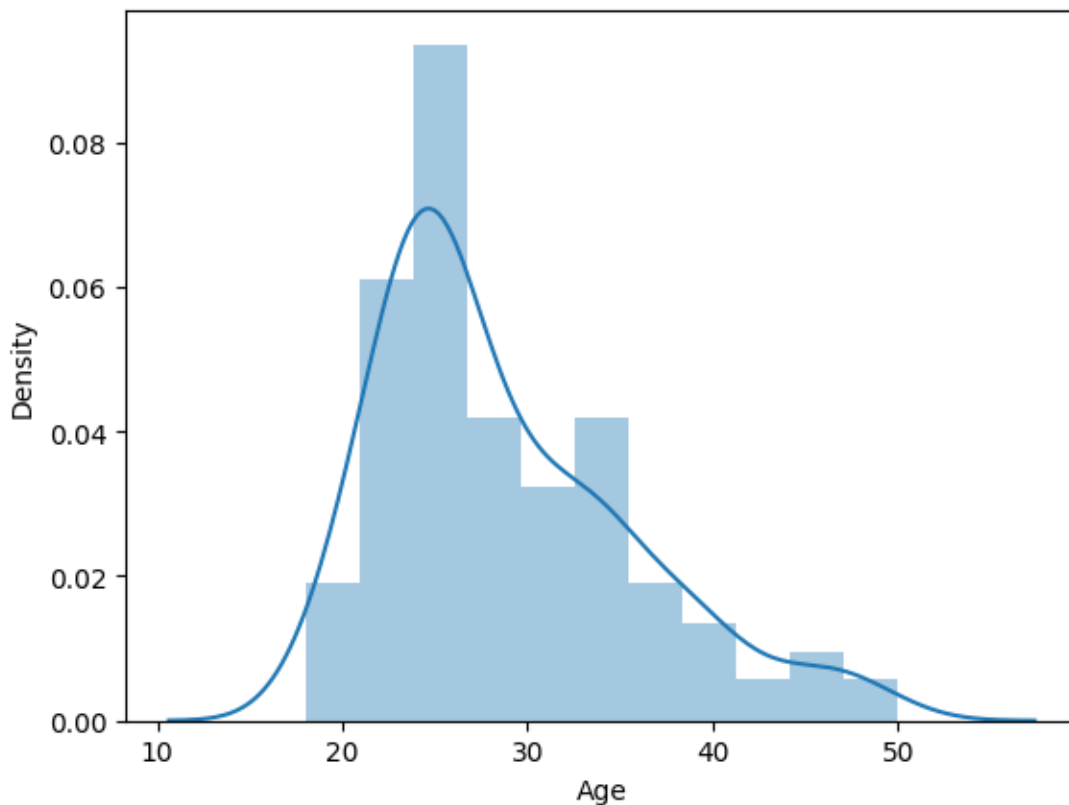
<ipython-input-13-29923ba59938>:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot(df['Age'])
```



4 Business Insights

Targeted Marketing: Design marketing campaigns specific to each age group. Social media and influencer marketing might resonate with younger demographics, while print or television ads might be more effective for older groups.

Product Development: Consider features and functionalities that cater to different age groups. Younger customers might value entertainment options like built-in TVs, while older customers might prioritize safety features like handrails or emergency stop buttons.

```
[14]: sns.distplot(df['Income'])  
      plt.show()
```

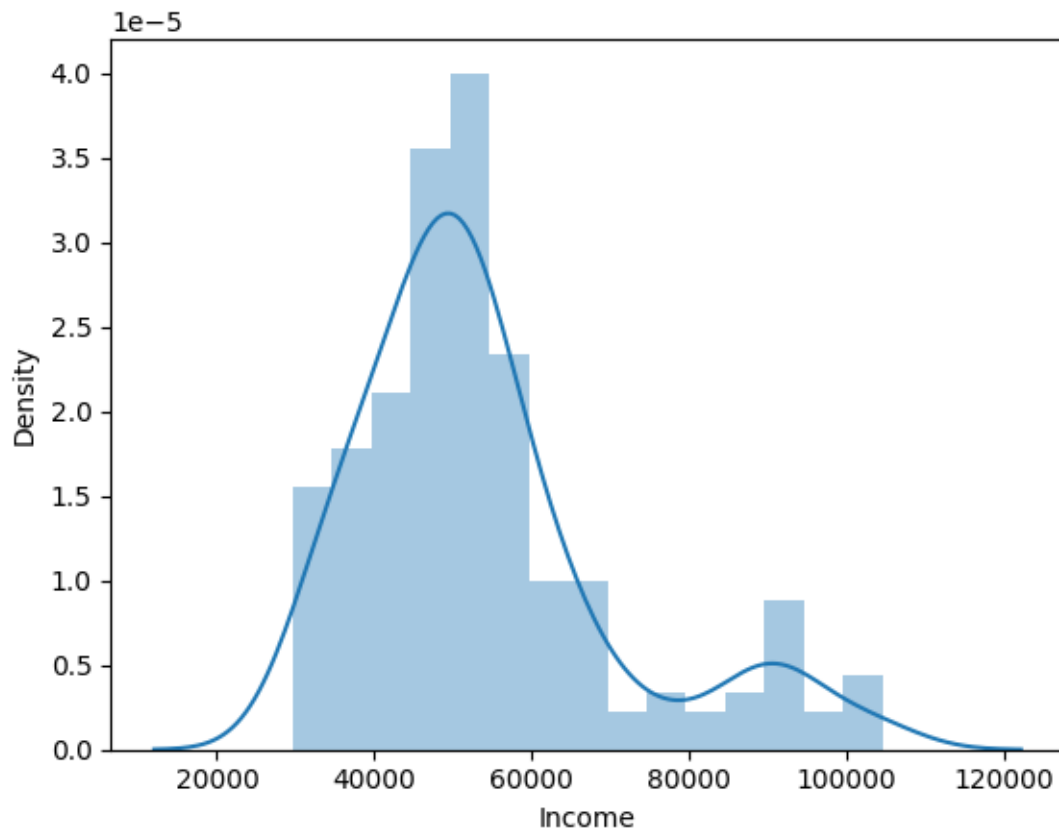
<ipython-input-14-50bd26e4ed8e>:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot(df['Income'])
```



5 Business Insights

Product Portfolio Strategy: Develop a range of treadmills (entry-level, mid-range, advanced) catering to different income segments and price points.

Targeted Marketing: Tailor marketing messages for each income segment. Highlight relevant features and value propositions:

High Income: Focus on technology, innovation, and premium experience.

Mid Income: Emphasize durability, warranty, and workout program variety.

Low Income: Promote affordability, value for money, and ease of use.

Upselling and Cross-selling:

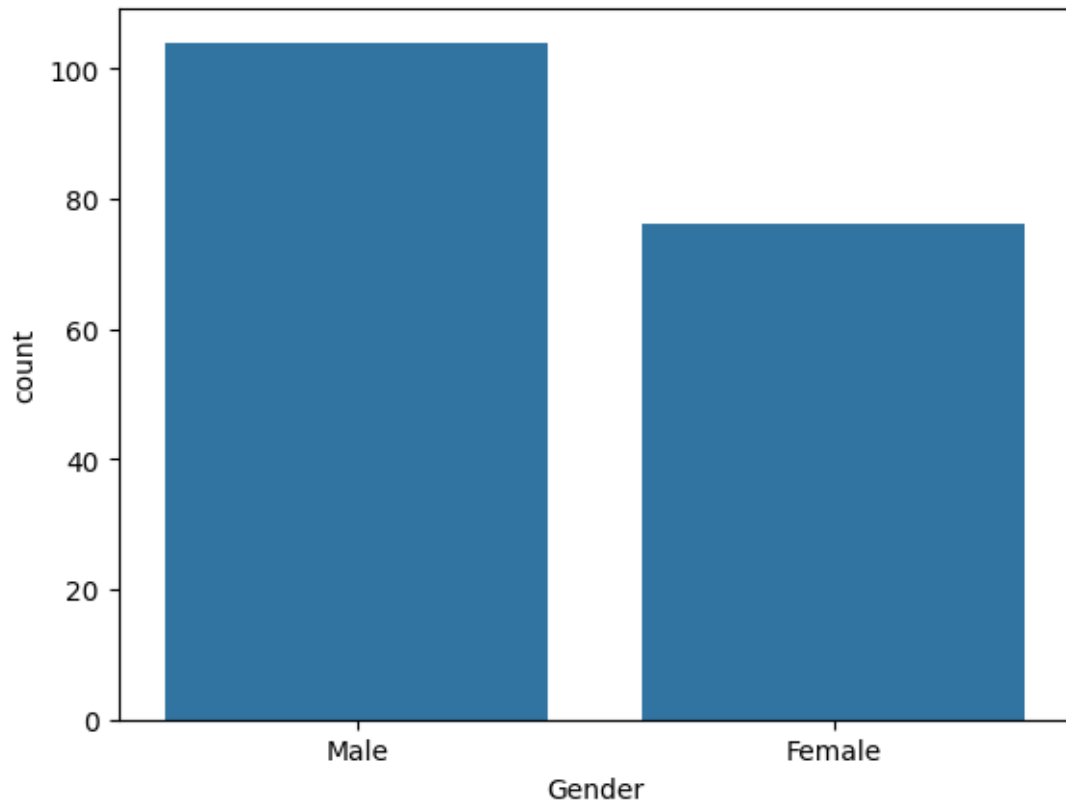
High Income: Offer premium accessories like heart rate monitors and fitness trackers.

Mid Income: Recommend yoga mats, resistance bands, and stretch bands.

Low Income: Suggest basic accessories like floor mats, water bottles, and workout towels.

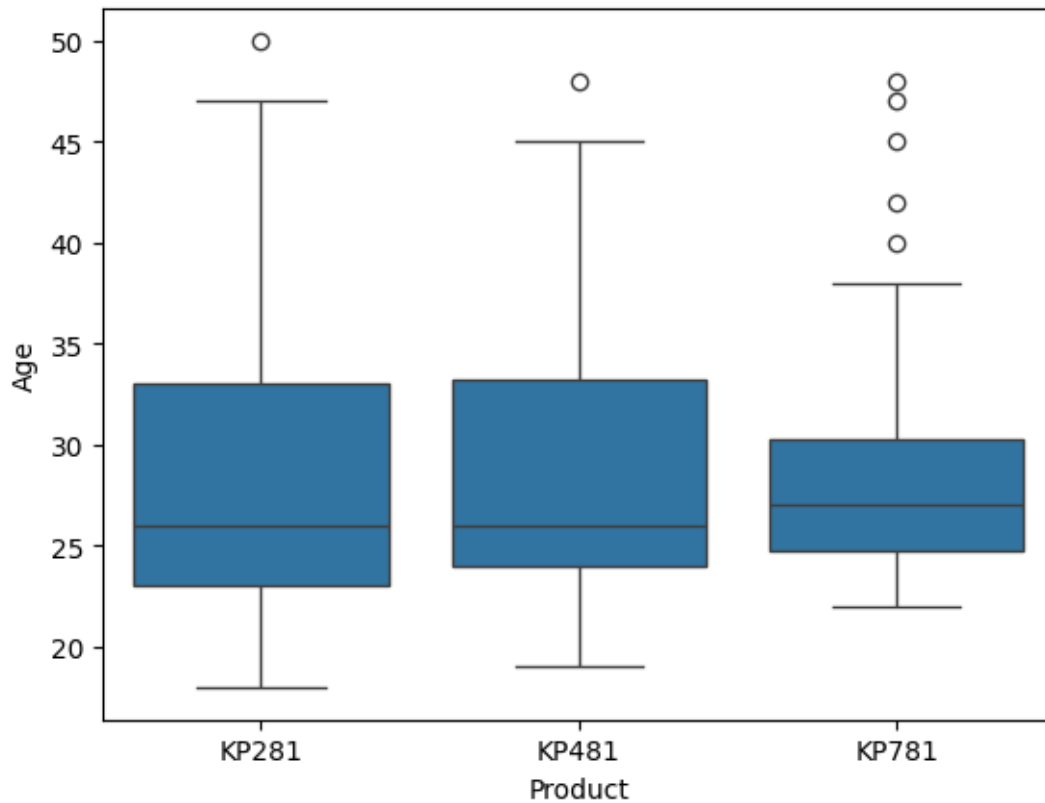
Financing Options: Consider offering financing plans to make high-end treadmills more accessible to budget-conscious customers.

```
[15]: sns.countplot(x='Gender', data=df)
      plt.show()
```



6 *Bivariate Visual Analysis*

```
[17]: # Boxplot for categorical variables against product purchased
sns.boxplot(x='Product', y='Age', data=df)
plt.show()
```



7 Business Insights

Here appears to be a spread of ages across all three treadmills.

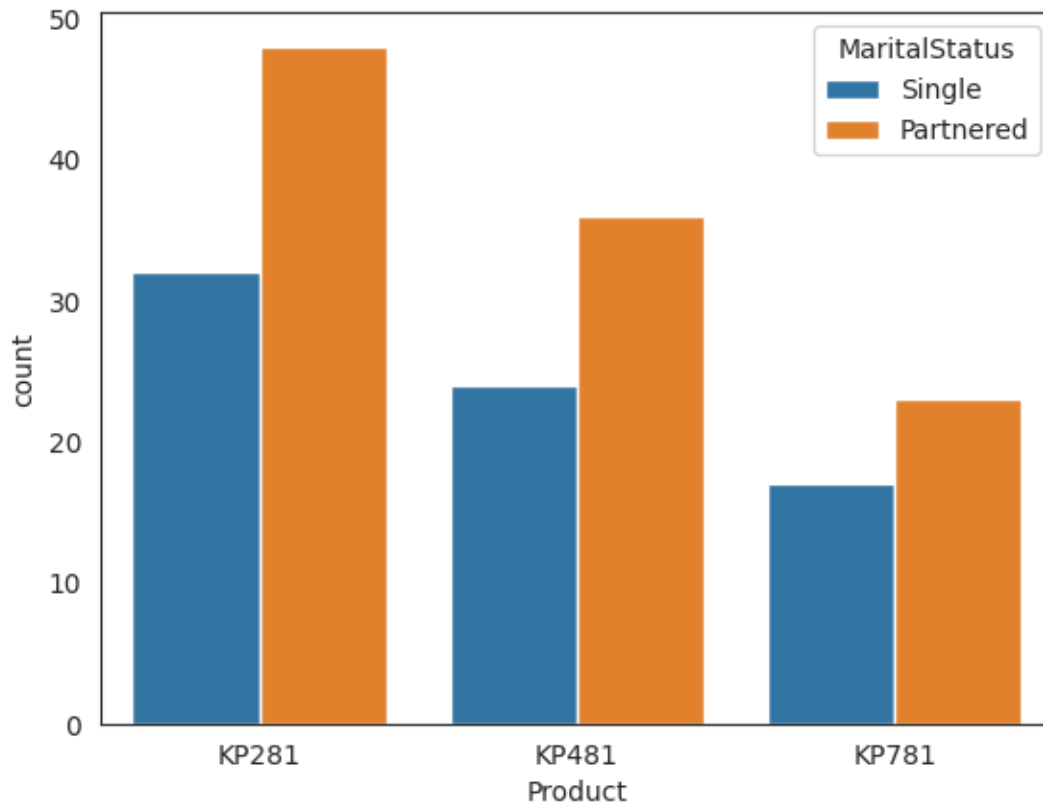
The boxplot for KP781 seems to show a tendencet towards a higher median age and potentially a wider range in ages compared to the other treadmills.

Roughly few customers with age above 30 use product KP781

Most of the customers are comfortable with KP281 product type

KP481 is the second highest popular product among the younger side of the customer

```
[30]: # Product purchased among Married/Partnered and Single
sns.countplot(data=df,x='Product',hue='MaritalStatus')
plt.show()
```



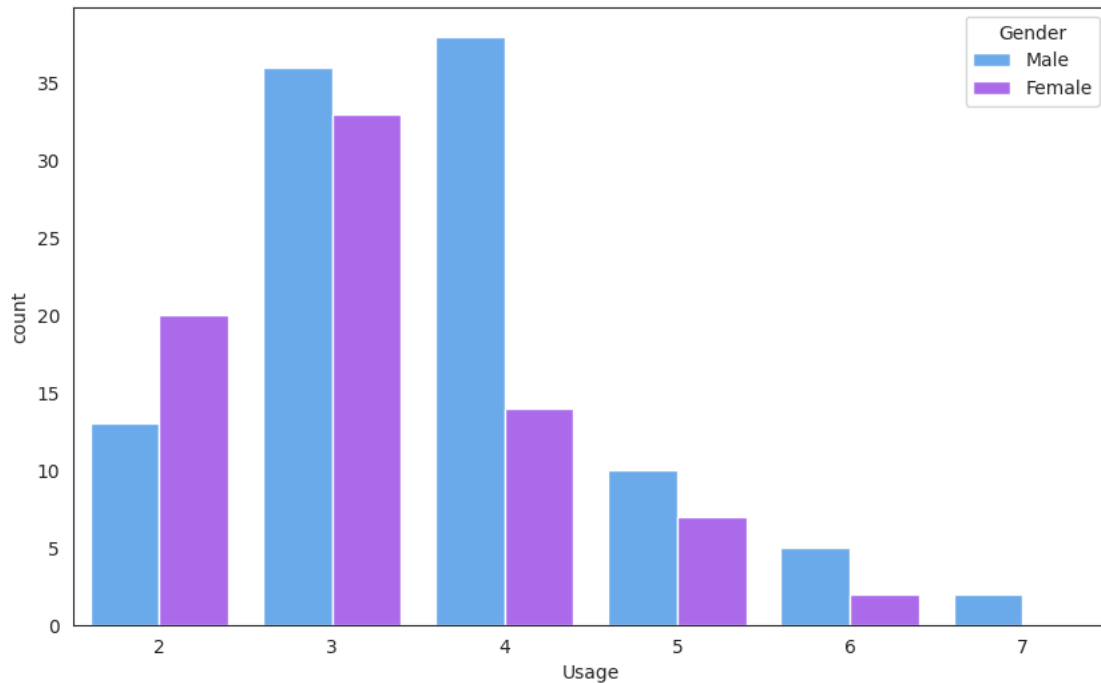
Business Insights

KP281 is the most preferred product among customers

KP481 is the second most preferred product among the customers

Between Singles and Partnered, Partnered customers are the major product purchasers

```
[31]: # Purchased product usage among Gender
plt.figure(figsize=(10,6))
sns.countplot(data=df,x='Usage',hue='Gender',palette='cool')
plt.show()
```



Business Insights

Among Male and Female genders, Male's usage is 4 days per week

Female customers mostly use 3 days per week

Only few Male customers use 7 days per week whereas female customer's maximum usage is only 6 days per week

```
[24]: attrs = ['Age', 'Education', 'Usage', 'Fitness', 'Income', 'Miles']
sns.set_style("white")
fig, axs = plt.subplots(nrows=2, ncols=3, figsize=(18, 12))
fig.subplots_adjust(top=1.2)
count = 0
for i in range(2):
    for j in range(3):
        sns.boxplot(data=df, x='Product', y=attrs[count], ax=axs[i,j],
                    palette='Set3')
        axs[i,j].set_title(f"Product vs {attrs[count]}", pad=12, fontsize=13)
        count += 1
```

<ipython-input-24-6aee39b5577c>:8: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(data=df, x='Product', y=attrs[count], ax=axes[i,j], palette='Set3')
<ipython-input-24-6aee39b5577c>:8: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(data=df, x='Product', y=attrs[count], ax=axes[i,j], palette='Set3')
<ipython-input-24-6aee39b5577c>:8: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(data=df, x='Product', y=attrs[count], ax=axes[i,j], palette='Set3')
<ipython-input-24-6aee39b5577c>:8: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

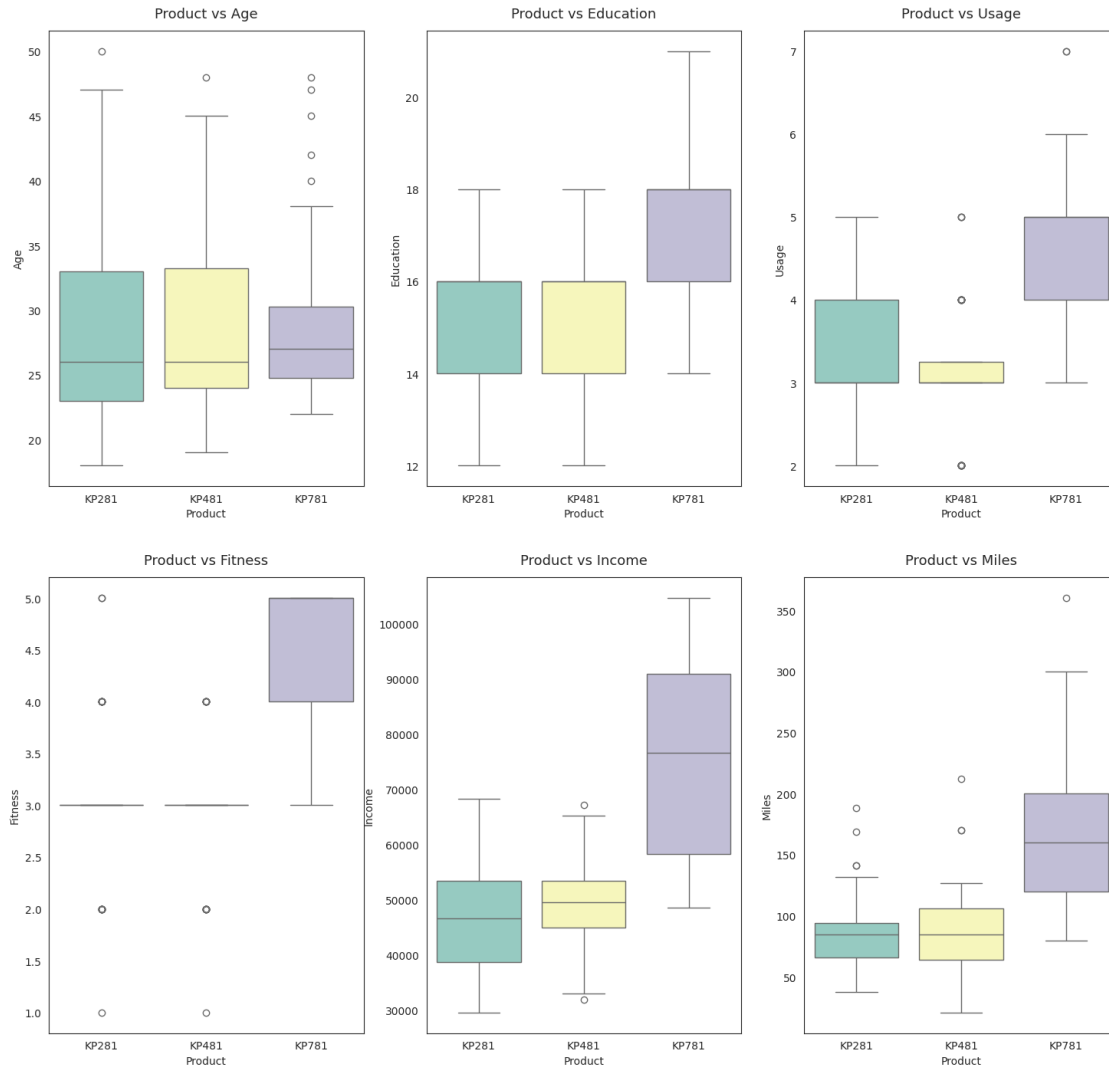
```
sns.boxplot(data=df, x='Product', y=attrs[count], ax=axes[i,j], palette='Set3')
<ipython-input-24-6aee39b5577c>:8: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(data=df, x='Product', y=attrs[count], ax=axes[i,j], palette='Set3')
<ipython-input-24-6aee39b5577c>:8: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(data=df, x='Product', y=attrs[count], ax=axes[i,j], palette='Set3')
```



Business Insights

Product vs Age

Customers purchasing products KP281 & KP481 are having same Age median value.

Customers whose age lies between 25-30, are more likely to buy KP781 product

Product vs Education

Customers whose Education is greater than 16, have more chances to purchase the KP781 product.

While the customers with Education less than 16 have equal chances of purchasing KP281 or KP481.

Product vs Usage

Customers who are planning to use the treadmill greater than 4 times a week, are more likely to purchase the KP781 product.

While the other customers are likely to purchasing KP281 or KP481.

Product vs Fitness

The more the customer is fit (fitness ≥ 3), higher the chances of the customer to purchase the KP781 product.

Product vs Income

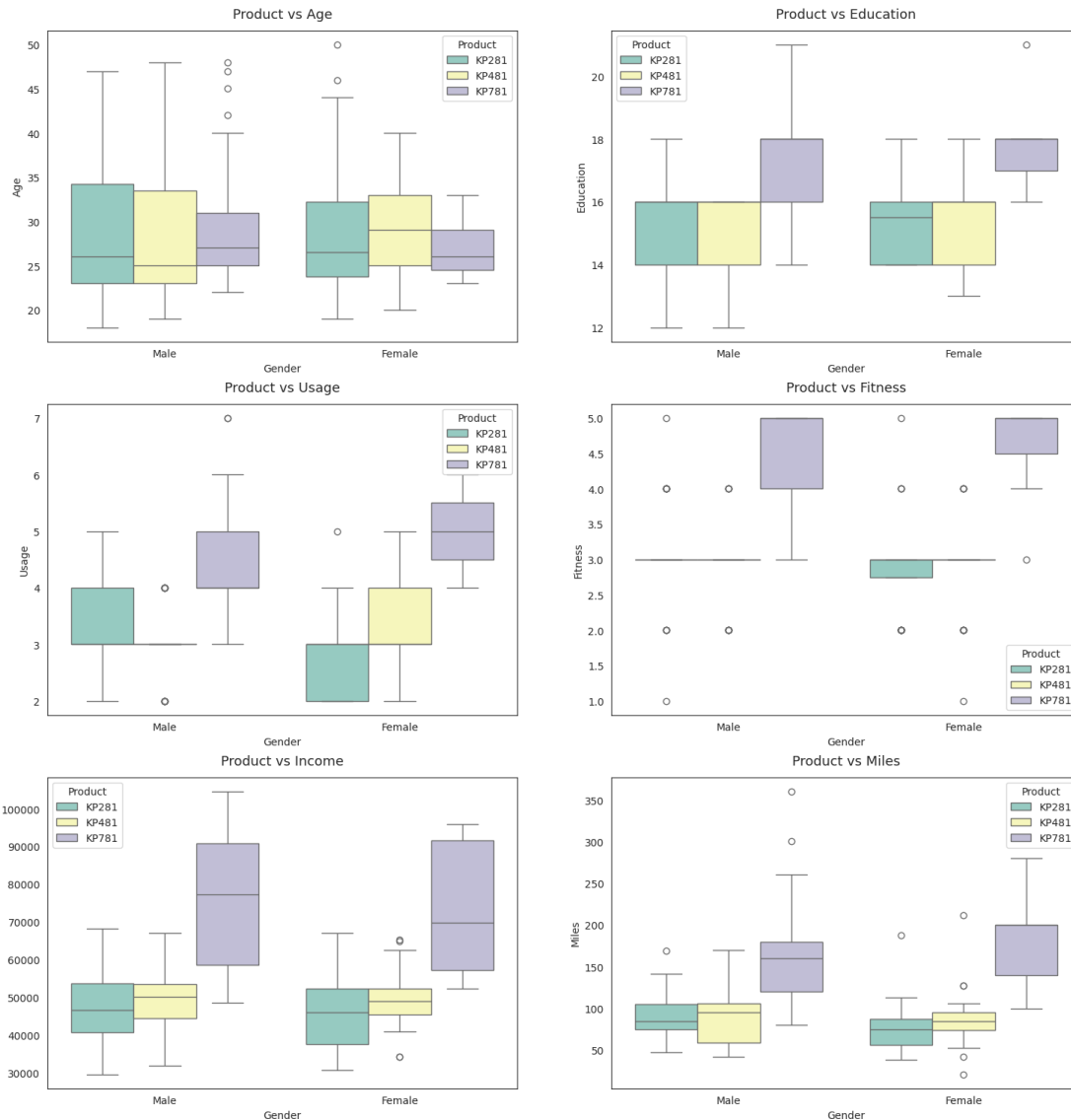
Higher the Income of the customer (Income ≥ 60000), higher the chances of the customer to purchase the KP781 product.

Product vs Miles

If the customer expects to walk/run greater than 120 Miles per week, it is more likely that the customer will buy KP781 product.

8 *Multivariate Analysis*

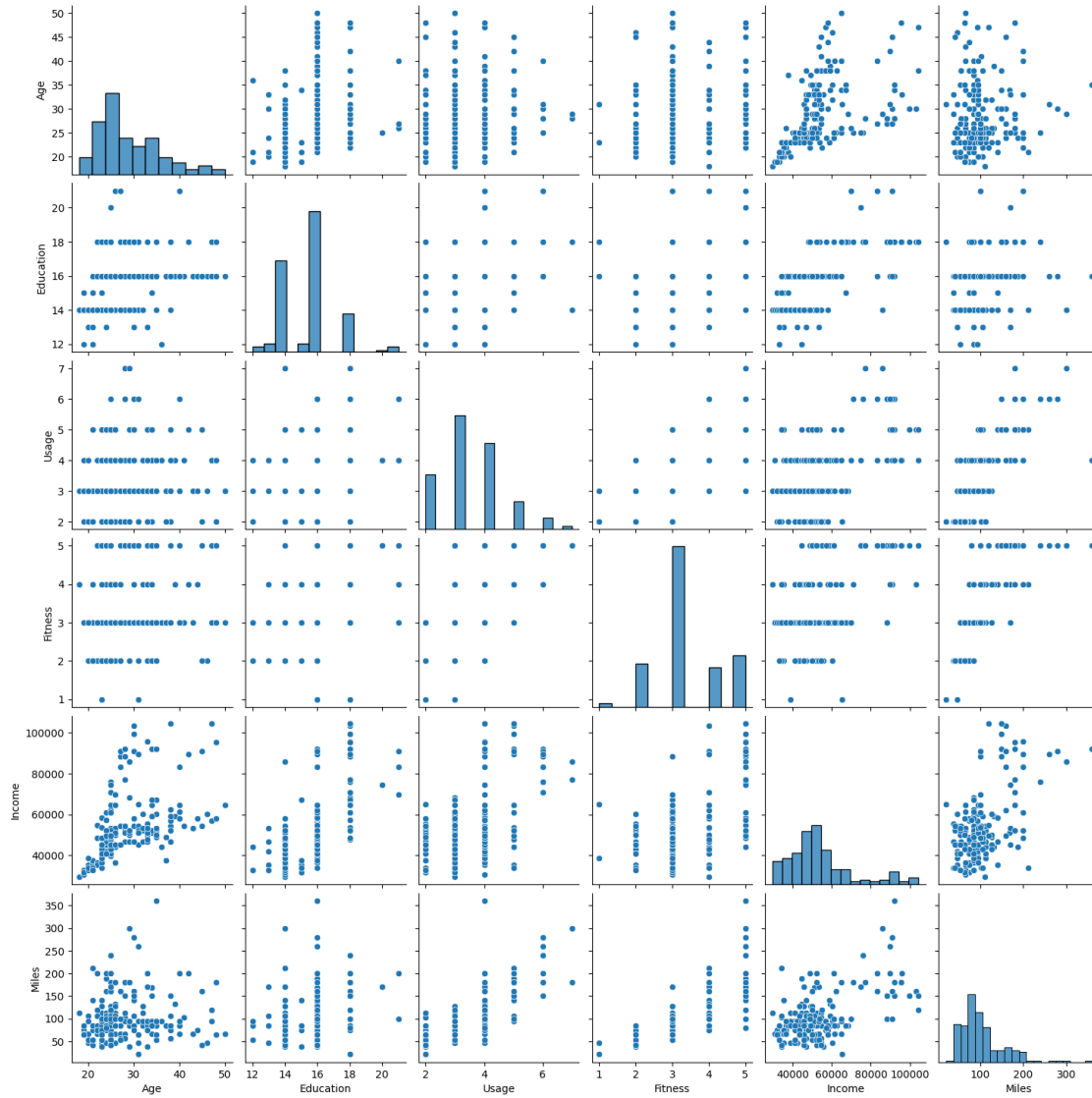
```
[25]: attrs = ['Age', 'Education', 'Usage', 'Fitness', 'Income', 'Miles']
sns.set_style("white")
fig, axs = plt.subplots(nrows=3, ncols=2, figsize=(18, 12))
fig.subplots_adjust(top=1.3)
count = 0
for i in range(3):
    for j in range(2):
        sns.boxplot(data=df, x='Gender', y=attrs[count], hue='Product',
                    ax=axs[i,j], palette='Set3')
        axs[i,j].set_title(f"Product vs {attrs[count]}", pad=12, fontsize=13)
        count += 1
```



Obervations

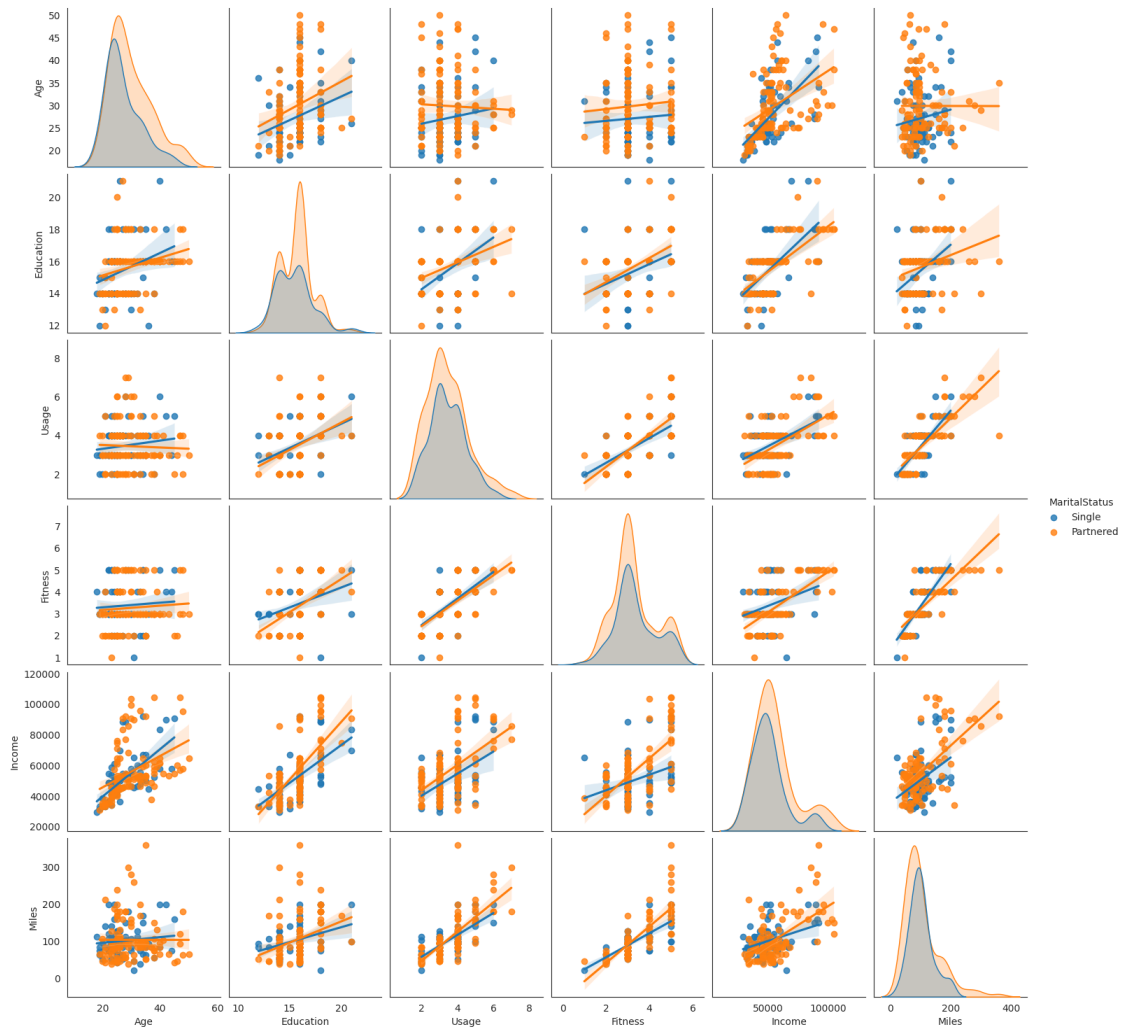
Females planning to use treadmill 3-4 times a week, are more likely to buy KP481 product

```
[19]: # Pairplot for correlation
sns.pairplot(df)
plt.show()
```



In the above pairplot the correlation with products and different attributes are as similar to previous observation

```
[29]: # Marital Status - pair plot
sns.pairplot(df,hue='MaritalStatus',kind='reg')
plt.show()
```



In the above pair plot the correlation with other attributes are pivoted around the marital status of the customer